CS 390 · Intro to Distributed Systems · Spring 2026

Concurrent Dispatch for Home Energy Systems

A Case Study in Structural Slack

A Go simulator of a solar home ends up arguing that Kubernetes HPA is doing the same thing as the battery policy that beats everything.

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The Dispatch Problem

Duke Energy NC — Schedule RSTC

Off-peak
$0.098 / kWh

On-peak
$0.224 / kWh

NEG (solar sell-back)
$0.030 / kWh

Peak window: Summer 3–6 PM · Non-summer 6–9 AM

Four Independent Actors

Solar panels

Variable producer — follows irradiance curve

Household load

HVAC + baseline + random appliance spikes

Battery

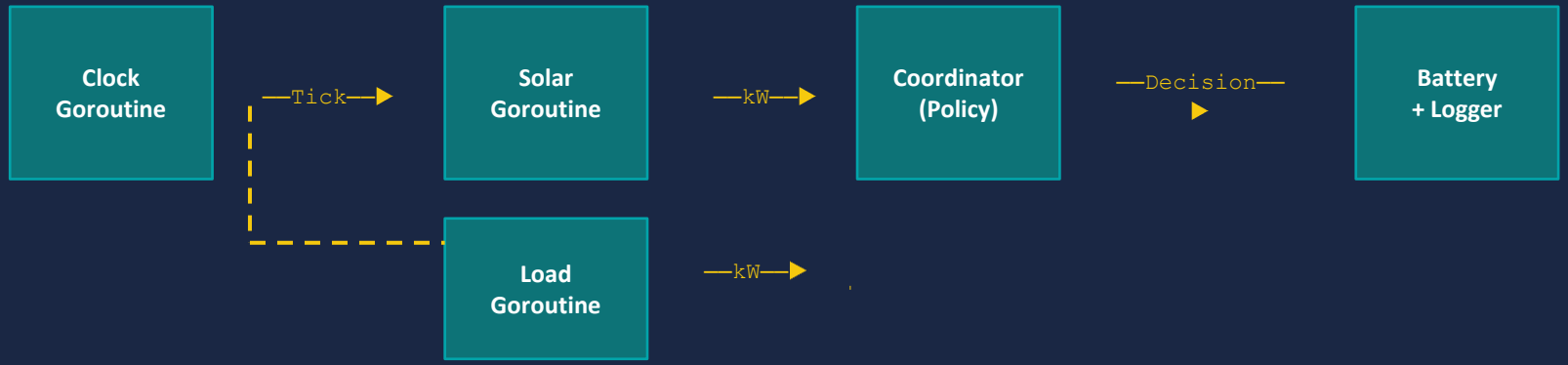
13.5 kWh / 5 kW continuous (Powerwall 2)

Grid

Buy at TOU rate · sell at NEG · or do nothing

Every 5 minutes, the home routes energy between these four actors — 288 decisions per day.

System Architecture — Goroutines & Channels



No Shared State

Components communicate only through typed buffered channels — never shared memory.

Logical Clock

The Tick carries logical time. Goroutines agree on time through messages, not a wall clock.

Real Control Pattern

Mirrors grid ISOs, SCADA, and cloud control planes: independent sensors, actuators, controller.

Reproducible by Design

Seeded RNG resets per policy — identical solar & load sequences guarantee a fair comparison.

Three Policies

All three are O(1) per tick — no forecasting, no state beyond current battery SOC.

Greedy Cheap-Fill

Grid-charge to 100% during every off-peak tick. Never discharge during on-peak.

Rationale: minimize the cost of energy entering the battery.

⚠ *Battery full by 1:20 AM → zero headroom for free solar. All surplus spills at $0.03/kWh.*

Greedy Discharge

Drain battery to 0% during on-peak. Even sends solar to grid when it could cover load.

Rationale: maximize kWh discharged during the most expensive period.

⚠ *Routes solar to grid at $0.03 instead of self-consuming at $0.224 — a 7.5× value loss.*

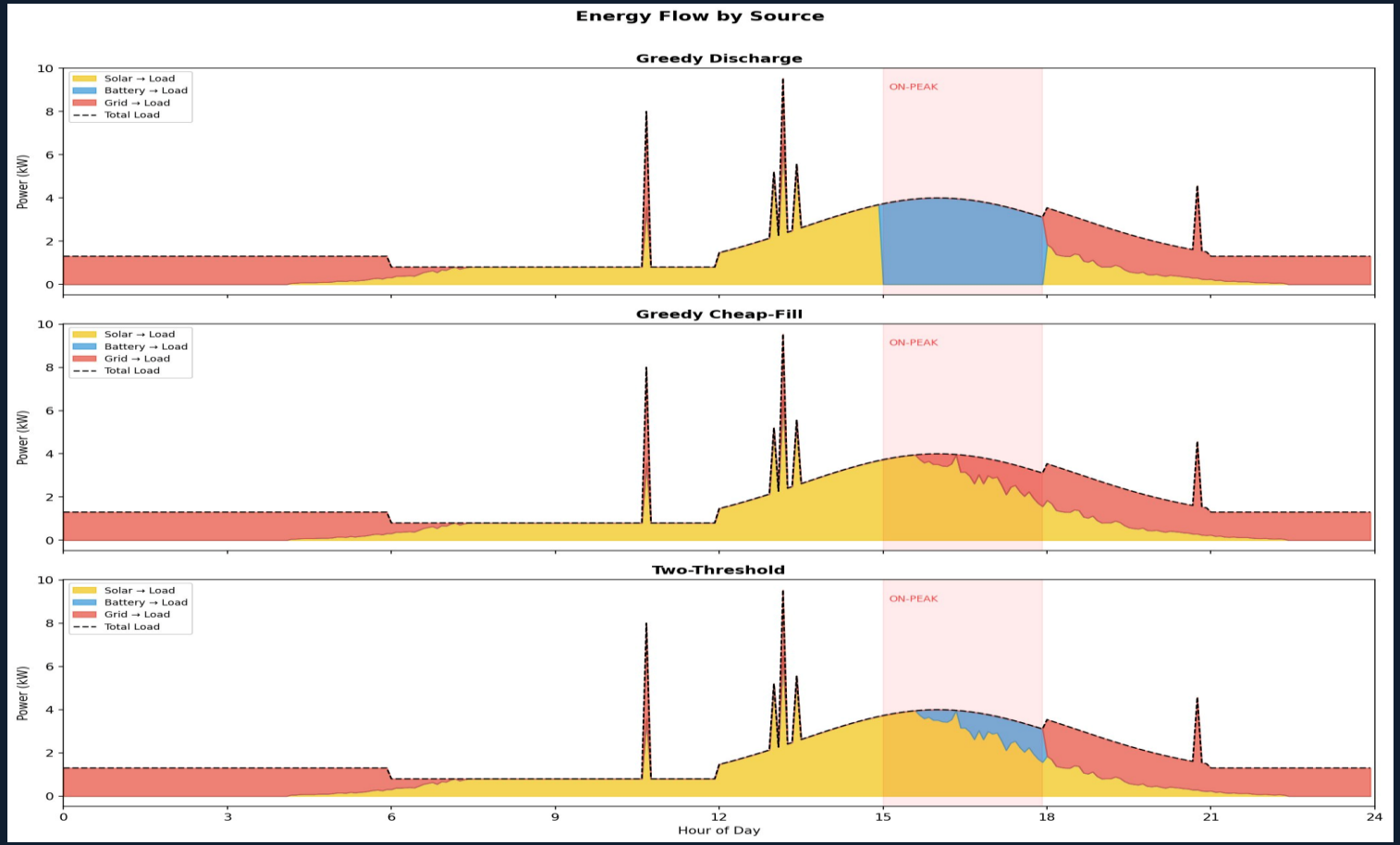
Two-Threshold ✓ Wins both seasons

Reserve floor X = 20% (2.7 kWh) · Grid-charge ceiling Y = 80% (10.8 kWh).

Solar can charge past Y (it's free). Grid charges only to Y — preserving headroom for solar.

✓ **Maintains 20% backup reserve · absorbs free solar in the 20% headroom gap each morning**

Summer Energy Flows — What Each Policy Actually Does



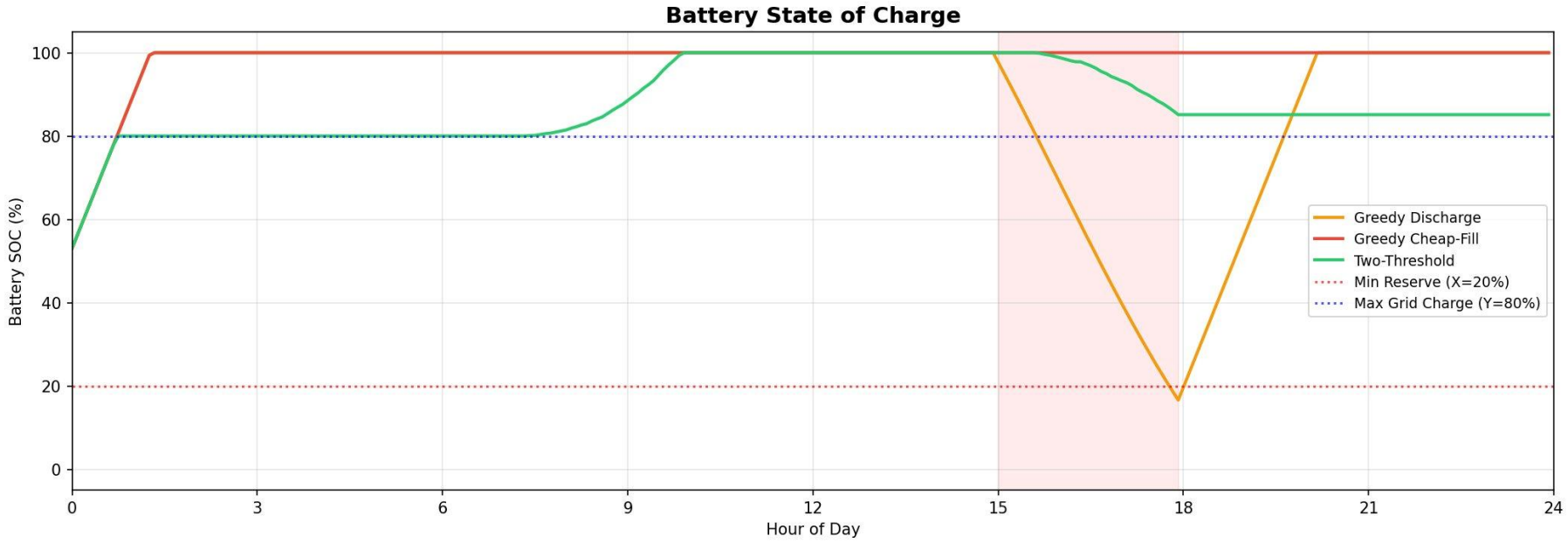
Solar → Load

Battery → Load

Grid → Load

Shaded = 3–6 PM on-peak

Battery State of Charge — Where Policies Diverge



Cheap-Fill

Hits 100% by 1 AM — no headroom for solar all day

Discharge

Drains to ~20% during on-peak — right at reserve floor

Two-Threshold

Stays within the 20–80% band; absorbs solar in headroom

The Greedy Ranking Flips Between Seasons

Season	Greedy Cheap-Fill	Greedy Discharge	Two-Threshold ✓
🌞 Summer	$2.26 (2nd)	**$2.63 (worst)**	**$1.62 (best)**
❄️ Non-summer	**$3.17 (worst)**	$2.30 (2nd)	**$2.03 (best)**

🌞 Summer: on-peak 3–6 PM overlaps solar tail

Cheap-fill's full battery has no room for free solar → spills at $0.03/kWh. Two-threshold's 20% headroom absorbs it instead.

❄️ Non-summer: on-peak 6–9 AM precedes solar

Discharge is now rational — battery covers the morning peak. But cheap-fill still refuses to discharge, paying peak rates regardless.

Two-threshold wins both seasons — by 28% and 36% over the next-best greedy policy.

Structural Slack as a Distributed-Systems Primitive

Home Energy

Cheap-fill: battery full at noon → free solar spills at $0.03/kWh

→

Cloud Analog

Over-provisioning — idle pre-scaled capacity, on-demand premium when real spikes arrive

Home Energy

Discharge: 0% SOC at end of peak → no reserve, no backup capacity

→

Cloud Analog

Aggressive scale-down — cold starts, pool exhaustion, cascading timeouts under load

Home Energy

Two-threshold: 20% floor + 80% ceiling, no forecasting needed

→

Cloud Analog

K8s HPA utilization band · AWS ASG MinSize / MaxSize · steady-state slack

"The cost of idle capacity is almost always less than the cost of not having it when you need it — whether the unit is a watt-hour or an instance-hour."

Thank You

Questions?

Source: github.com/djmurong/energy-systems · Go 1.21+ · Duke Energy Schedule RSTC · NREL TMY3 · Tesla Powerwall 2